



Course Number:	RAIK 371H		
Course Title:	Management Science Principles		
Credit Hours:	3		
Space and Time:	110 KAUF	Tue, Thu	9:30–10:45 AM

Instructor: Dr. Seth Polsley
Office: 145 KAUF
E-Mail: spolsley3@unl.edu
Office Hours: Walk-in: 145 KAUF Tue 12:30–2:00 PM
Wed 2:30–3:30 PM
Thu 11:00 AM–12:00 PM
Appointment: Book online via go.unl.edu/polsley-calendar

Teaching Assistants: Elena Belachenko: ebelashchenko2@huskers.unl.edu

This is the final course in the sequence of Raikes School's data and modeling curriculum. It explores modern computational aspects of management science, with an emphasis on technology and its impact on different business sectors. We will cover a variety of topics from an engineering perspective, including programming implementations of various algorithms, while also diving into industry-focused discussions and case studies examining the roles of these technologies today. We will also consider traditional topics related to advanced machine learning, predictive analytics, and management science.

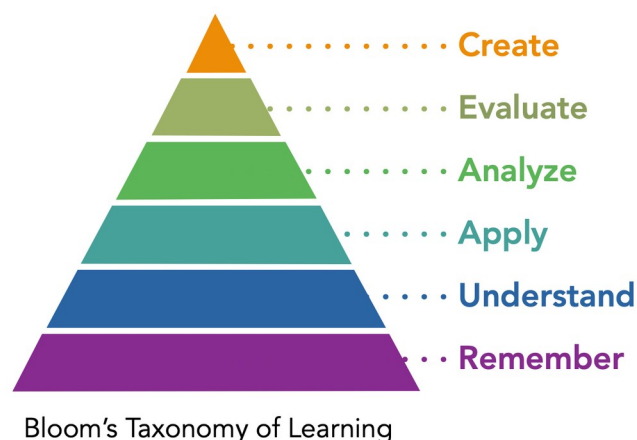
Specific concepts time series analysis; prediction; spreadsheet-based analysis; and optimization; as well as coverage of many different applied machine learning approaches, including ensemble learning, as they relate to different industries like recommender systems, blockchain, text processing, and more.

- RAIK 370

As this is the final course in the core Raikes School data science curriculum, there is an emphasis on preparing students for Design Studio and real-world work. In addition to covering underlying algorithms, the class will look at many different applications of data science today and practice effective communication of core concepts through a course project.

- Recognize essential theoretical concepts for data analysis, classification, prediction, and optimization as related to various industries.
- Understand the underlying operations of and differences between multiple algorithms like neural networks, recommender systems, blockchain, simulation, and others.

- Apply class concepts to practical problems through a business lens.
- Clearly communicate in written and oral formats to technical and general audiences, interweaving management perspectives with technical knowledge and details of implementations and analyses.
- Work with a team to develop a project focused on a solution to an identified market, with a technically-grounded implementation, and corresponding sharing of work and results.



Textbook and Resources

Canvas

Canvas will be the hub for class materials, assignment descriptions, and grades.

1. To access the course, go to canvas.unl.edu
2. Log in with your UNL account
3. Click on "HNRS:DATA&MODLS III:FOUND RAIK371H"

Gradescope

Certain assignments will utilize Gradescope due to its advanced grading capabilities. These include certain worksheets, quizzes, and homeworks.

All tools and resources will be linked through Canvas, and any other relevant ones will be shared with the class as applicable.

Grading

Scale

This class will use the plus-minus scale for final grades; depending on the distribution, a curve may be applied. The instructor reserves the right to determine final grade assignments near a borderline, based largely on the student's perceived effort and participation.

A+	A	A-	B+	B	B-	C+	C	C-	D+	D	D-	F
≥ 98	≥ 94	≥ 90	≥ 87	≥ 83	≥ 80	≥ 77	≥ 73	≥ 70	≥ 67	≥ 63	≥ 60	< 60

Components

The class is primarily structured according to a formative approach, using a collection of elements worth comparatively few points that accumulate over the course of the semester.

Category	Percentage	Breakdown
Activities	25%	15% class activities, 10% weekly discussions
Assignments	25%	15% homeworks (3), 10% case study discussions
Assessments	20%	6% group quizzes, 4% self-learning reflections, 10% exam
Application	30%	30% project

- **Activities:**

Class Activities consist of in-class discussions, prompts, and problem sets used to practice concepts through active-learning.

Weekly Discussions, aka “Boom or Doom”s, will be hosted on Canvas forums. Each week, a different group (separate from project teams) will select a topic centered around disruptive technology in an industry sector and write a short post describing the technology, arguing why it is generally a more positive or negative application, and open the discussion for others to respond. This grade includes both creating a discussion once in the semester and engaging with others’ posts throughout the term.

The lowest 2 within each of the “Activities” sub-categories will be dropped.

- **Assignments:**

There will be 3 **Homeworks** through which you can practice applied aspects of the course. In particular, CS-oriented content is more likely to be emphasized in the homeworks. A pool of 2 “late days” is available to give you some flexibility for late assignments, as described below.

The **Case Study Discussions** portion will be a segment of the class around early March in which small teams will put together a short position presentation, with backing material, explaining their stance on a particular topic or case, such as responsible AI or an example of industry disruption (e.g., Zoom overtaking Skype during the pandemic). This component is weighted similarly to the exam but more focused on the management science aspect of the course. In some ways, it will be reminiscent to a Boom or Doom but more about communicating the idea to the class.

- **Assessments:**

Quizzes will be given as a means of practicing core concepts and will generally be held on a weekly timetable, excluding other events such as the case study discussions. They will be given as group quizzes and may be done open-book, open-note. The goal of quizzes is to help you both review concepts and to practice working through problems as augmented by peer discussion.

Self-learning Reflections are a form of individual assessment that will be given a few times through the semester, probably every other week or so. These will be short activities to be completed on your own that will entail a few conceptual questions as well as reflection-based activities such as designing and answering your own questions. The goal here is to have a few formative points in the assessment’s category that are individual to you.

There will only be a single **Exam** in the class. Expect the exam to include conceptual and applied problems largely in the vein of content covered on the quizzes. There is no final as the course shifts towards completion of the project, and the final exam slot, if used, will be for project presentations.

- **Application:**

The group **Project** is the single largest component of the course. It will cover ideation, development milestones, and culminate in a final presentation at the end of the semester in place of a final exam.

Part of the goal of this class project is to take everything you have learned in the data and modeling curriculum and synthesize a solution to a demonstrable problem and will incorporate aspects of management science relevant from across Raikes School curriculum.

Deadlines

Assignments are typically due at 11:59pm of the given deadline day, unless otherwise stated, and are pursuant to the make-up and late work policies described below.

Class Policies

Attendance:

Although it is not a standalone component, attendance is embedded into the grade via short participation exercises. Between the facts that participation exercises may not be featured in every class and that the lowest 2 are dropped, you should have some flexibility to miss class. In truth, it may be possible to regularly miss without a significant grade impact, but I do strongly encourage class attendance as it is an important way to keep up in the course and is among the “perceived effort and participation” guidance used when making evaluations as described in the “Grading” section.

Absences and Make-up Work:

University-sponsored events and extreme personal circumstances may be considered as excused absences and eligible for make-up work. See the Student Affairs Student Absence Policy & Notifications (<https://studentadvocacy.unl.edu/student-absence-policy-notifications>) rule for more information. Notably: “a student should recognize that not every course can accommodate excused absences, and neither absence nor notification of an absence, relieves them from meeting the course requirements.”

While there is some built-in flexibility through dropped points, there may be times when you could miss key deadlines or events like quizzes and exams. In case of an absence or need to make-up work, please notify the instructor as soon as possible prior to an absence and provide documentation of excused absences within 3 days of the absence.

Generally, make-up work for excused absences cannot be accepted more than 30 days after the deadline. Unexcused exam absences will result in a 0. The instructor reserves the right to individually assess each case based on the circumstances as pertaining to excusing or adjusting the penalty of late work that has exceeded the typical 5-day or extended 30-day window.

The key takeaway regarding absences or missed work is to keep me and the TAs informed to the extent possible. Double check that assignment submissions have all your files, use dropped points judiciously, and please reach out if you are worried about falling behind!

Late Submissions:

Typically, assignments are due at 11:59 PM on the day of the deadline, although most assignments accept late work. By default, late work receives a deduction of 0.5% (a half-point) per hour with up to 12% per day deducted. After 5 days, late work will receive no credit.

In alignment with my intent to include flexibility in each aspect of the class, everyone has 2 “freebie” deadline extension that can be used to push a deadline by a 24-hr period. Simply email me, with the TA cc’d, stating that you would like to extend the deadline for the assignment in question, and we’ll update it as soon as possible, no explanation needed! For practical purposes, this is really only useful for Assignments portion of the class but could give you an extra day if a particular homework needs it!

Instructional Continuity Plan:

If an in-person class meeting is prevented due to weather conditions or other circumstances, the class will generally be held online over Zoom instead. In such an event, the corresponding Zoom link will be shared via email and linked on Canvas prior to class.

Accommodations:

If you will need an accommodation in this course, you may register with the Services for Students with Disabilities (SSD) at <https://www.unl.edu/ssd/> to submit your request. Please also feel free to visit go.unl.edu/polsley-calendar to book an appointment with me. I would like to make the class as accessible as possible and am happy to discuss with you the best plan for meeting your needs.

Use of Large Language Models and Generative AI Tools Policy:

In this course, students are permitted to use large language models (LLMs) and other generative AI tools to assist with their assignments, so long as they follow the guidelines below.

- **Disclosure:** Students must disclose any usage of LLMs or generative AI tools in their assignments by including a supplementary document that explains what tool was used, the extent of its usage, and how it helped them learn in relation to the assignment outcomes. The supplement should be included with the submission of the assignment.
- **Original Work:** Students are expected to use LLMs and generative AI tools as aids to their own original work. Submitting work that is entirely or primarily generated by an LLM or other AI tool without proper citation and disclosure is considered academic dishonesty and will be subject to the appropriate consequences.
- **Learning Outcomes:** Students should use LLMs and generative AI tools in a way that supports their learning and understanding of the course material. The use of these tools should not replace the student's own critical thinking, analysis, or problem-solving skills.
- **Responsible Use:** Students are expected to use LLMs and generative AI tools responsibly and ethically. This means avoiding any usage that could potentially undermine the learning objectives of the course or create unfair advantages for themselves or others.

By adopting this policy, we aim to promote a culture of transparency, accountability, and responsible use of technology in support of student learning. We encourage students to take advantage of these powerful tools while also developing their own skills and knowledge.

Note that the above policy was written using llama.cpp with the llama-70b-chat model using the following prompt in order to demonstrate usage of generative tools in this course:

“I’m designing a course syllabus for a college class, and I want to add a policy regarding the usage of large language models and generative AI tools like ChatGPT. I would like a policy that allows students to use such tools, so long as they include a supplement explaining what they used, the extent of usage, and how it helped them learn in regards to the assignment outcomes. For instance, in this case, I'd say I used the Llama 70b model, include this prompt, and mention how using an LLM to generate this portion of the syllabus helps illustrate the policy I'd like to adopt. Can you help write that policy for me?”

Raikes School Values:

Remember the Raikes School values (<https://raikes.unl.edu/raikes-school-core-values/>): intellectual curiosity, humility, gratitude, excellence, accountability, resilience, and community. As Raikes School students, you are expected to adhere to these values. You should be honest with your work, courteous to and understanding of others when collaborating, and curious and accountable in the classroom.

University Policies

Students are responsible for knowing the university policies and resources found on the page at <https://go.unl.edu/coursepolicies>.

Tentative Schedule

Week	Topic(s)
1	Machine learning review
2	Generative AI and industry contextualization
3	Optimization
4	More optimization
5	Time series and forecasting
6	Auto-regression and autocorrelation
7	Markov models
8	Case study discussions: responsible AI, industry disruption, etc.
9	
10	Special topics: language processing, block chain, cybersecurity, swarm AI, reinforcement learning, etc.
11	
12	Exam – Thursday, Apr. 9
13	Project work / remaining topics
14	Final Presentations
15	Final Presentations
16	Final Presentations – Thursday, May 7, 10:00 AM— 12:00 PM

